

Question

Sample Program: Write a Java program to check whether the given number is Prime Palindrome or not. CO1.

Algorithm

1. Start the program.
2. Read a number n from the user.
3. Check whether the given number is prime.
4. To check prime, divide the number by numbers from 2 to $n/2$.
5. If the number is divisible by any number, it is not prime.
6. Store the original number in a variable.
7. Reverse the digits of the number.
8. Compare the reversed number with the original number.
9. If the number is both prime and equal to its reverse, display "**Prime Palindrome**".
10. Otherwise, display "**Not a Prime Palindrome**".
11. Stop the program.

Steps for Execution

1. Open a Java-supported IDE or command prompt.
2. Create a Java file named **PrimePalindrome.java**.
3. Type or paste the given program into the file.
4. Save the file.
5. Compile the program using:

```
javac PrimePalindrome.java
```

6. Run the program using:

```
java PrimePalindrome
```

7. Enter a number when prompted.
8. The program checks whether the number is prime and palindrome.
9. The result is displayed on the screen.

UML Class Diagram

```

+-----+
| PrimePalindrome |
+-----+
|               |
+-----+
| + main(String[] args) : void |
+-----+

```

OUTPUT:

```

PS C:\Users\HP\Documents\CSMD 247> & 'C:\Users\HP\AppData\Local\Programs\Eclipse Adoptium\jdk-21.0.1
2.8-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Roamin
g\Code\User\workspaceStorage\a865700a6b715bbf5e59fb88dd8f06f9\redhat.java\jdt_ws\CSMD 247_e2e0bfb\bin
' 'Primepalindrome'
Enter a number: 23
23 is not a Prime Palindrome.
PS C:\Users\HP\Documents\CSMD 247>

```